香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

DDA2001: Introduction to Data Science

Lecture 5: Random Variable

Shuang Li

Recap

Random Variable (RV)

- A mathematical formalization of a quantity or object which depends on the outcome of the random experiment.

Example: Flip a fair dice.

- If >3, you win $2.
- If ≤ 3, you lose $1.

Random experiment outcome: $\omega \in S,\ S = \{1,2,3,4,5,6\}$
Random variable: $X(\omega) \in R,\ R = \{1,2\}$

The set of different values that a RV can assume is called its range/ support/sample space.

PMF

- For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$. A probability mass function $f(\cdot)$ is a function such that:

✓ $f(x_i) \geq 0$ for all $x_1, x_2, \dots, x_n$.

✓ $\sum_{i=1}^{n} f(x_i) = 1$

✓ $f(x_i) = \boxed{P(X = x_i)}$ for all $x_1, x_2, \dots, x_n$.

Probability that the random variable takes value x_i.

Expectation

- Motivation: how to quantify return?
- Play it N rounds, the average earning.

$$\frac{\sum_{\omega\in S} X(\omega)N(\omega)}{N}$$

- Taking limits $N \to \infty$

$$\sum_{\omega\in S} X(\omega)P(\omega)$$

- Let the range of X be R: $\sum_{\mathrm{x}\in \mathrm{R}} \mathrm{x}\ \mathrm{P}(X = x)$

Mathematically, we call it the expectation or mean of X.
In notation, write it as E[X]

CDF

- P(X≤ x): called the **cumulative distribution function** (CDF) of X.

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

Note: In this course, we focus on single-variate random variable, namely the outcome of an experiment can always be represented by a single number.

CDF

For example: Consider the probability of some "bad events" using a Cumulative Distribution Function (CDF):

- **Bad event 1:** The random variable is smaller than a certain value. **Bad event 2:** You lose more than a specific amount of money.
- **Bad event 3:** A machine stops working if fewer than three of its components are operational.

CDF

- In each of these cases, we can use the CDF to calculate the probability of these "bad events" happening, by looking at the probability that the random variable is less than or equal to a certain value (for example, the amount of money lost or the number of operational components).

Variance

A quantity that can be used to measure the risk

Example: Flip a fair dice.

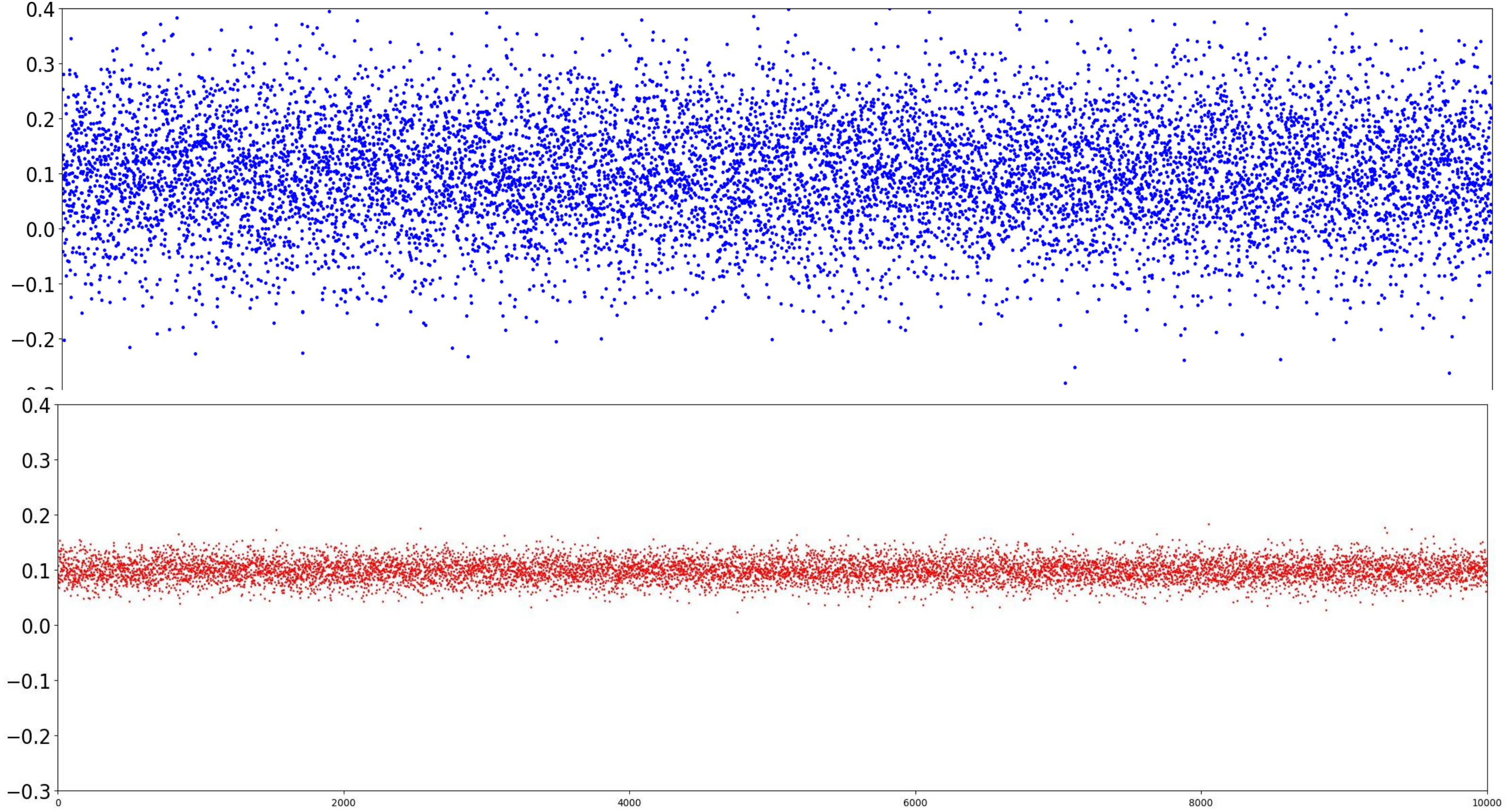
Game 1:

- If > 3, win \$2.
- If ≤ 3, lose \$1.

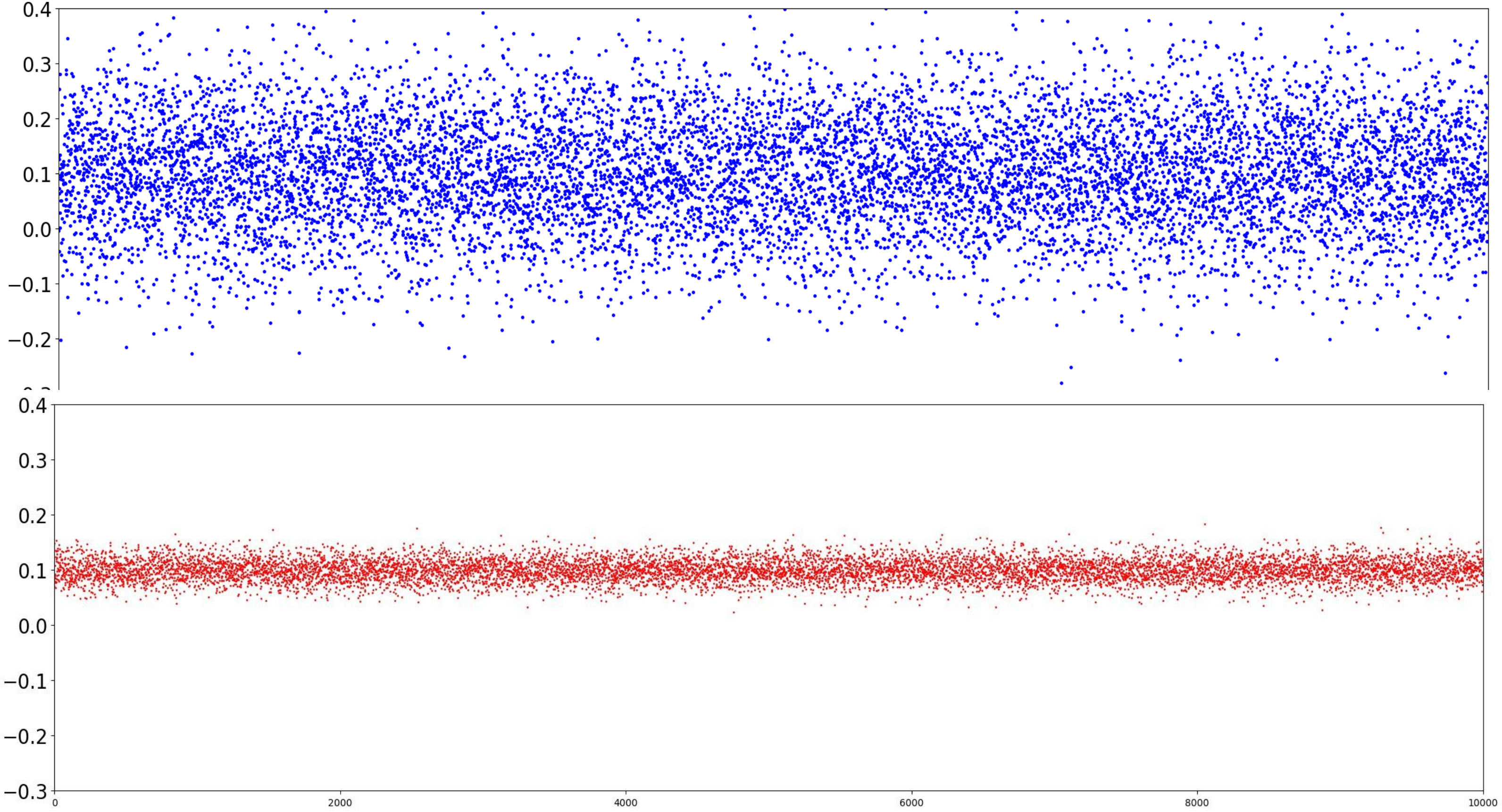
Game 2:

- If > 3, win \$10 0000 0003.
- If ≤ 3, lose \$10 0000 0001.

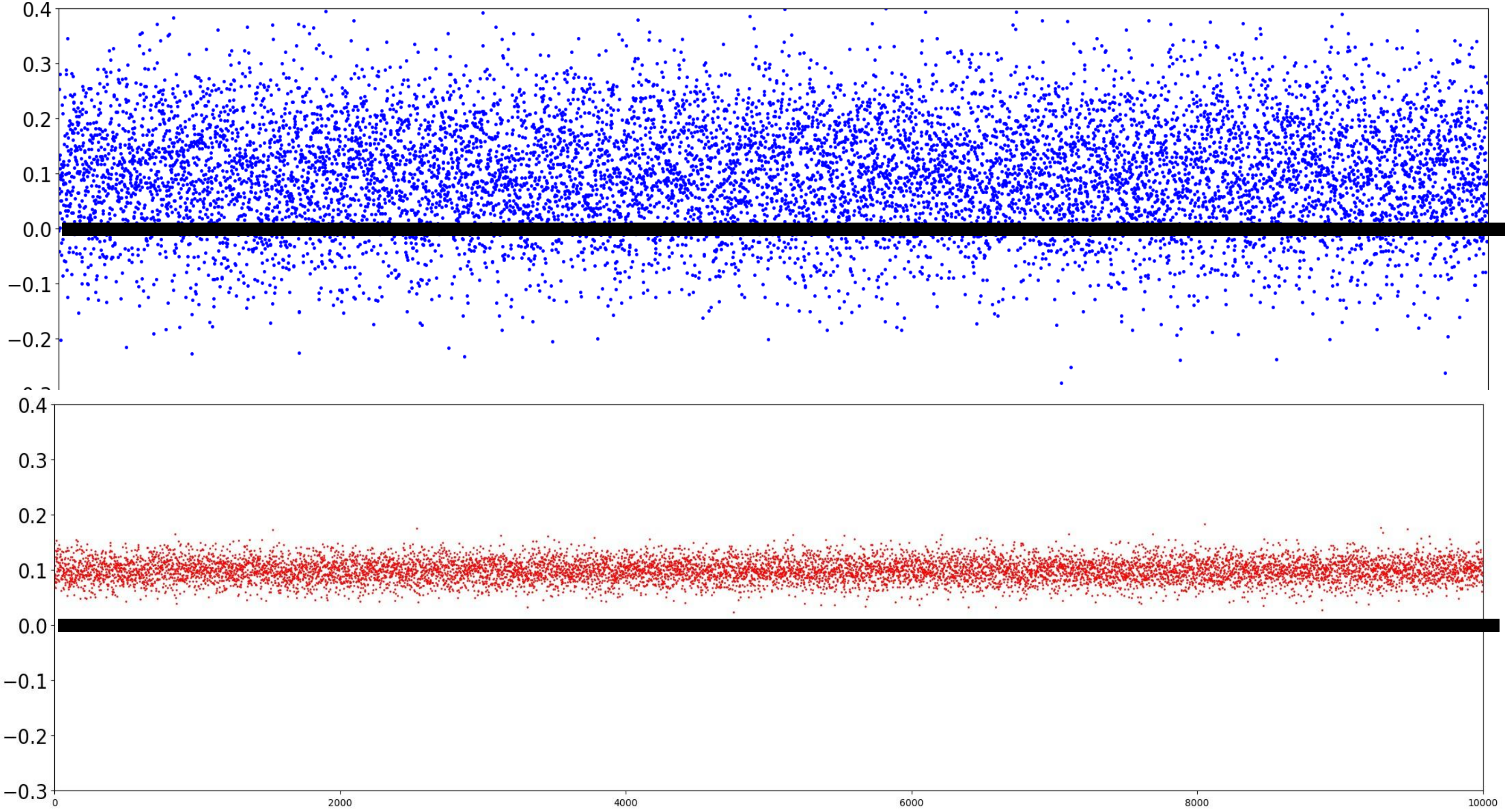
- Suppose I track the returns of two stocks 10,000 times. Below is my record



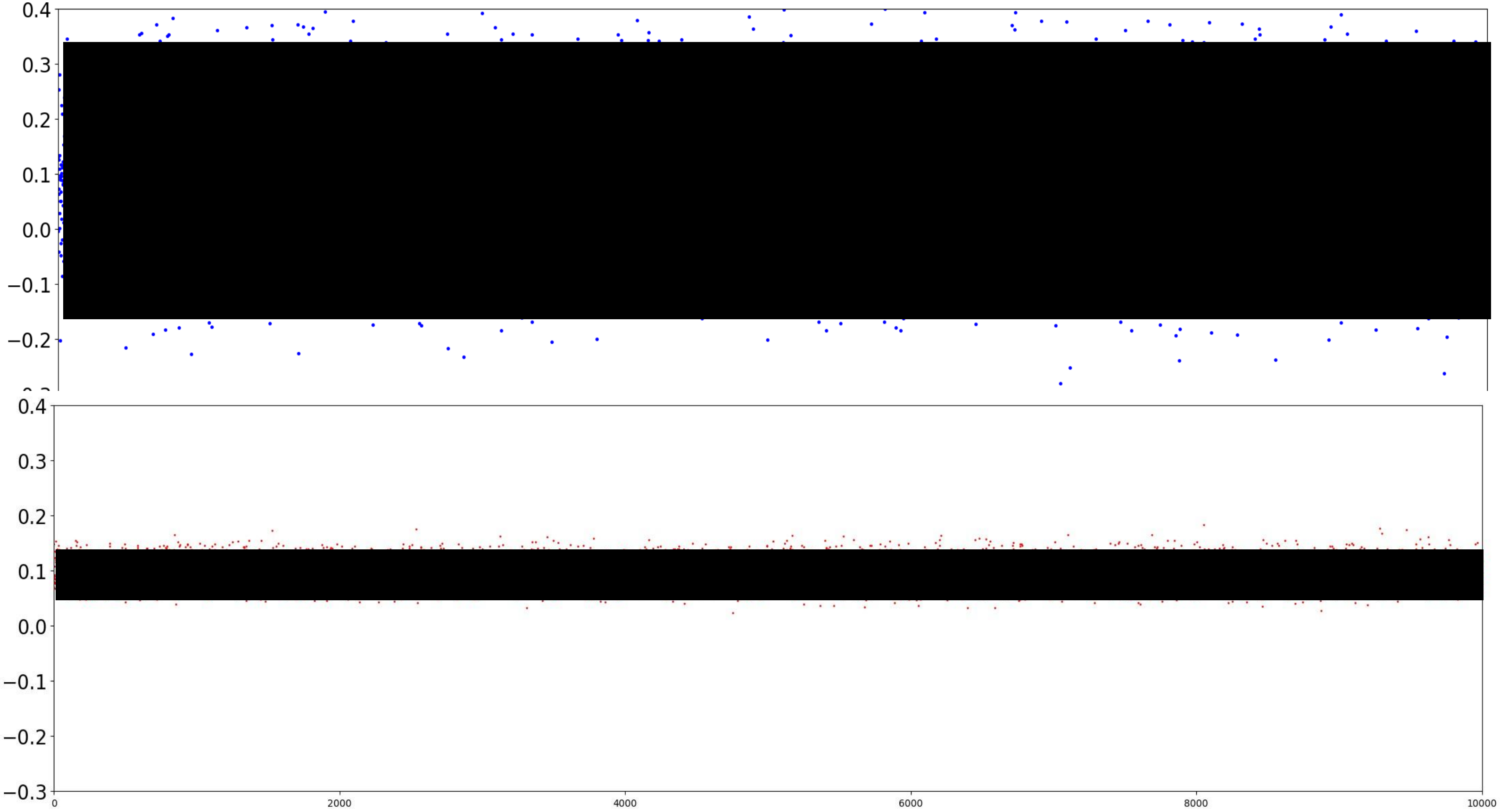
- The average return of each stock is 0.1
- Which stock will you invest in?



- More likely, investing in a Blue stock will cause loss



- Risk: fluctuation.



How far is a random variable from its mean, on average?

$$|X - E[X]|$$

Note

- E[X] is a number while X is random
- |X- E[X]| is a RV

How far is a random variable from its mean, on $\boxed{\text{average?}}$

$$E[\,|X - E[X]|\,]$$

Note: this quantity equals to $\sum_x |x - E[X]|\, P(x)$

1. First calculate E[X]
2. Then calculate the above quantity

How far is a random variable from its mean, on average?

$$(X - E[X])^2$$

$$\mathrm{E}[(X - E[X])^2]$$

Note: this quantity equals to $\sum_x (x - E[X])^2 \, P(x)$

Mathematically, we call the above quantity the variance of X

Example: Investment

Maximize average return?

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No! Also reduce the chance of losing too much after one investment.

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Risk

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Real objective:

Example: Investment

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Real objective:

Maximize: Average Return - Risk

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Real objective:

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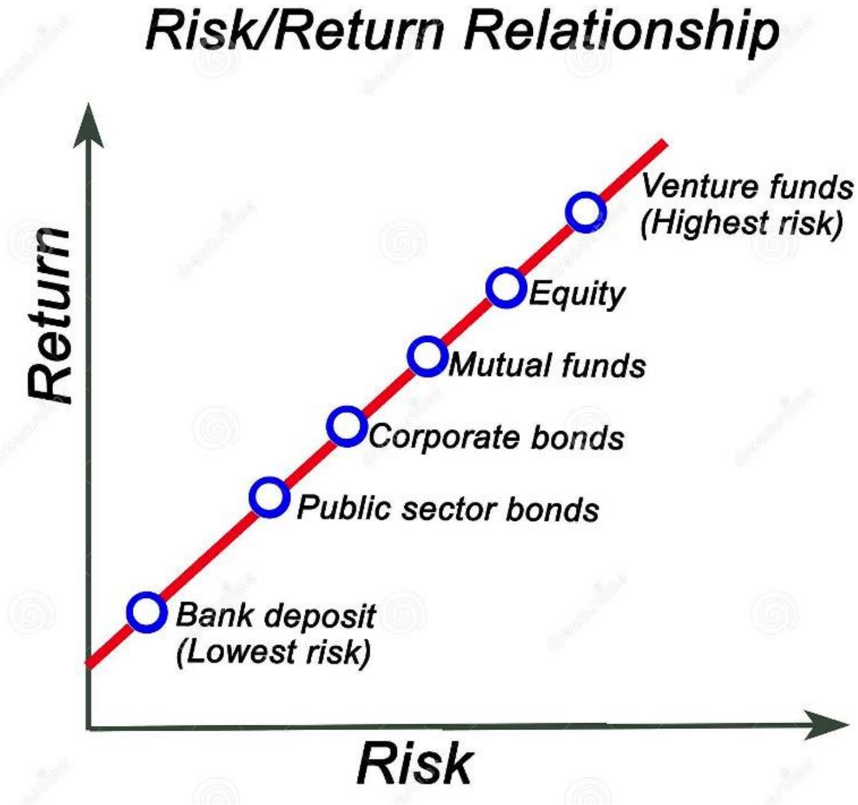
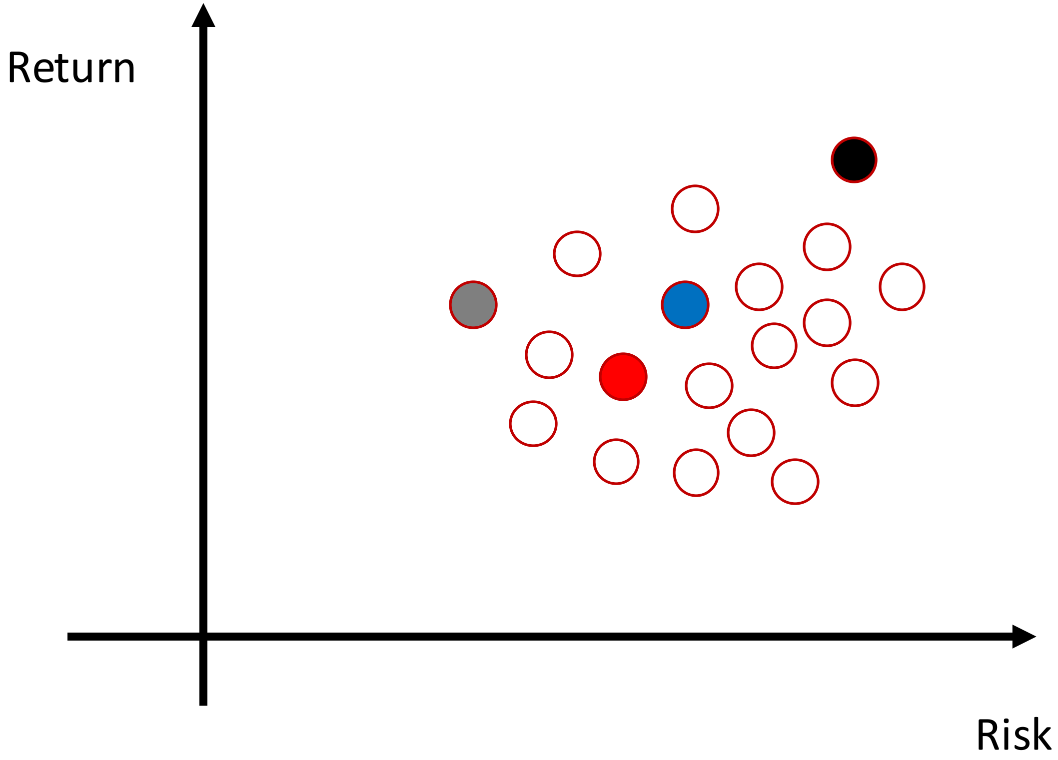
Markowitz Model

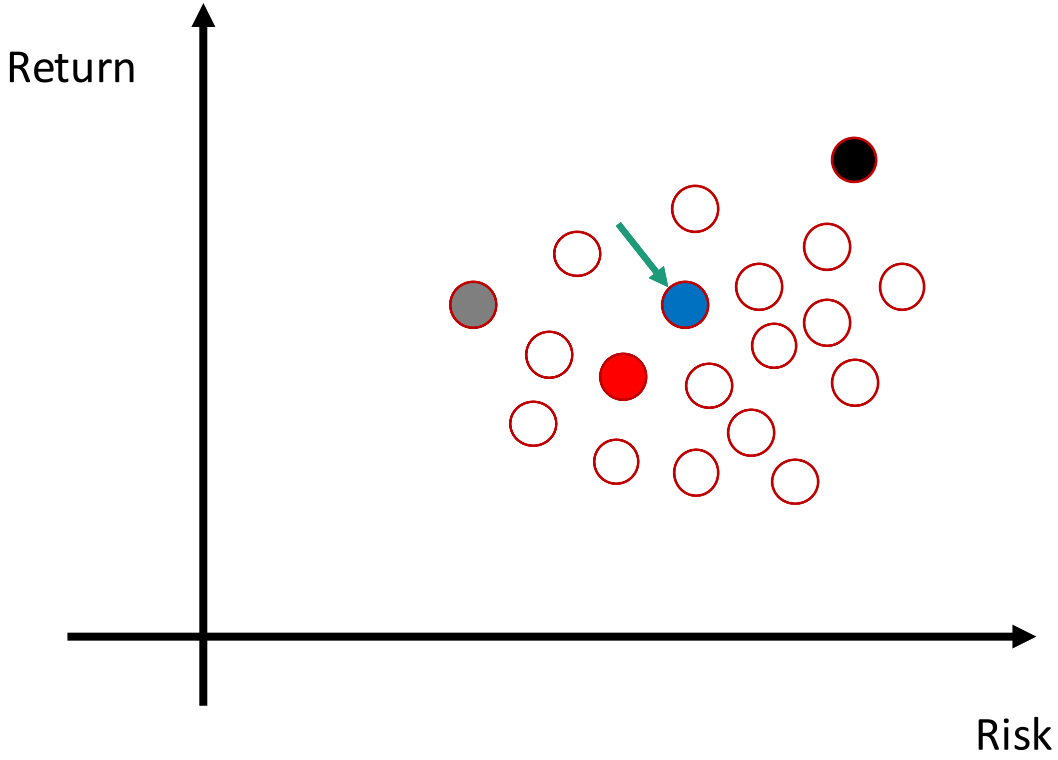
The first model you need to learn to work in finance.

Example: Investment



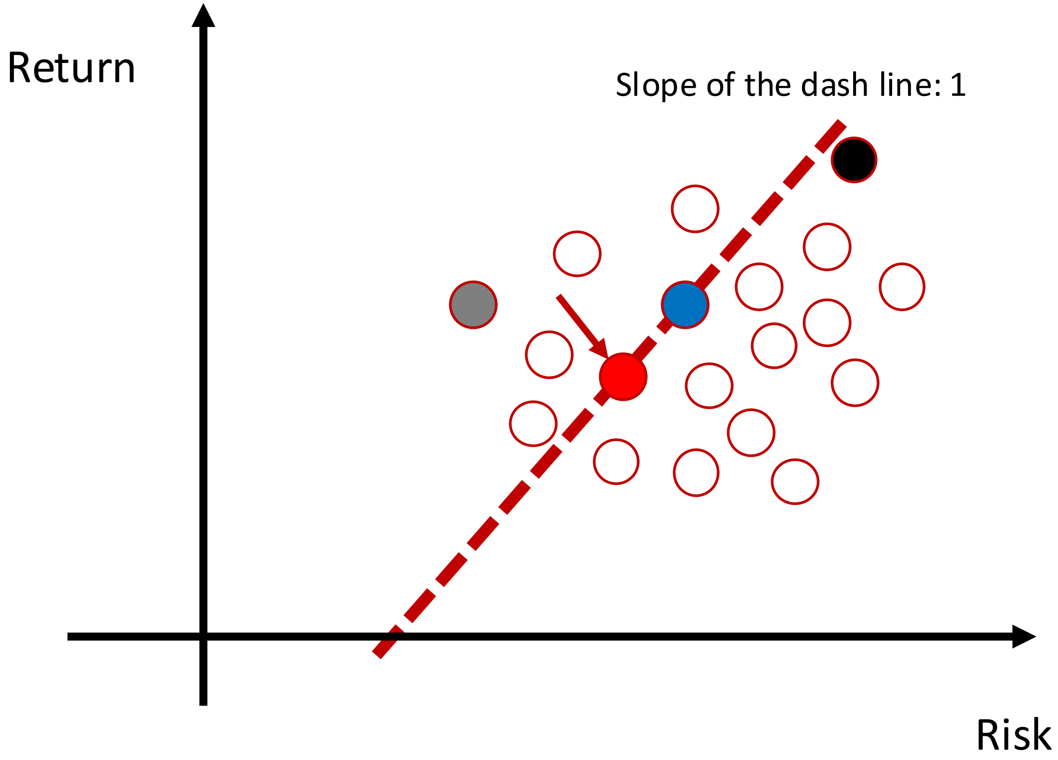


Each dot represents a stock



Objective: Return – Risk
Suppose you choose blue

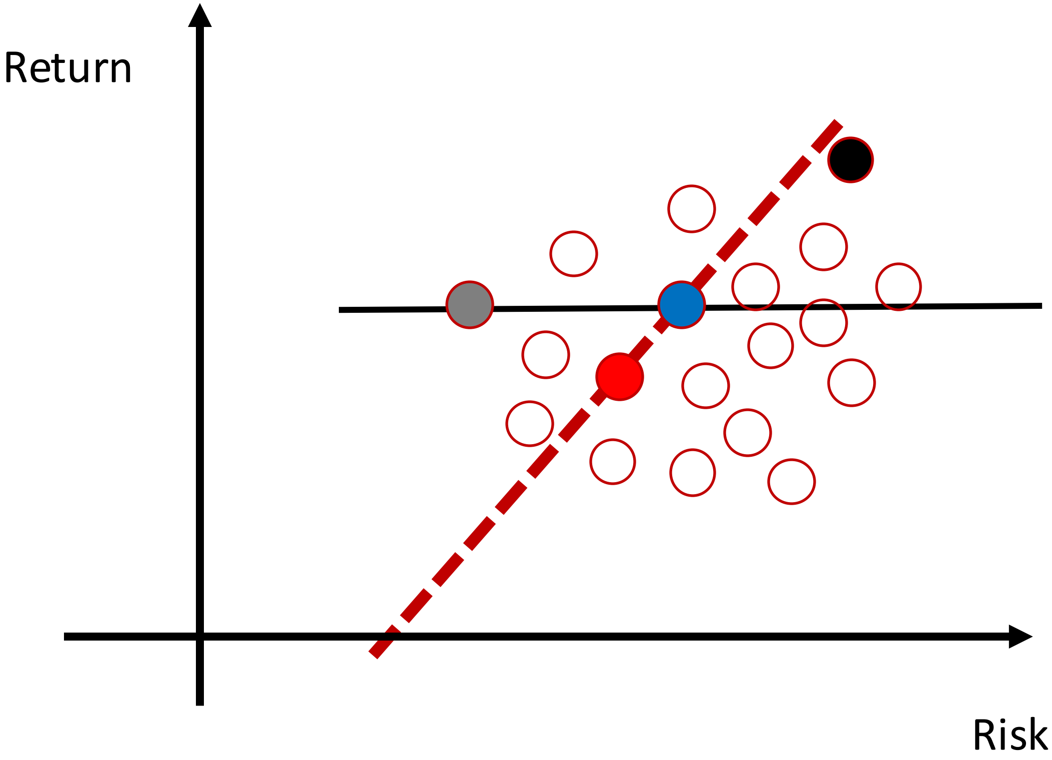
Each dot represents a stock



Objective: Return – Risk
Suppose you choose blue

Objective unchanged when choosing red

Each dot represents a stock



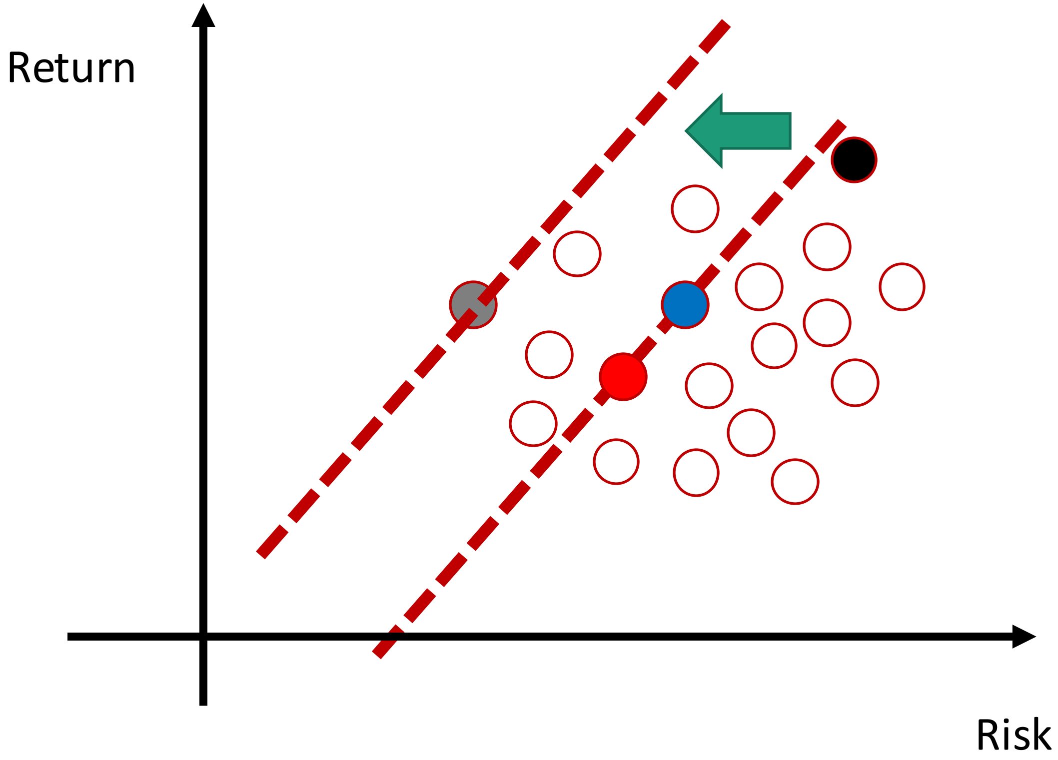
Objective: Return – Risk

Suppose you choose blue

When choosing grey:

same return but smaller risk

Each dot represents a stock



Objective: Return – Risk
Suppose you choose blue

When choosing grey:
same return but smaller risk

Best choice: move the line with slope 1 to left side until there is no dot at the left side of the line

Each dot represents a stock

Mean and Variance

- Mean

$$\mathrm{E}[X] = \Sigma_x x\, P(X = x) = \Sigma_x x\, f(x)$$

- Variance

$$\mathrm{Var}[X] = \Sigma_x (x - \mathrm{E}[X])^2\, f(x)$$

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- Toss a coin three times and count the number of heads.
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- What is $\mathrm{Var}[X]$?

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- What is $E[X]$?
- $E[X] = 0 * f(0) + 1 * f(1) + 2 * f(2) + 3 * f(3) = \frac{3}{2}$
- What is $\mathrm{Var}[X]$?
- $\mathrm{Var}[X] = \left(0 - \frac{3}{2}\right)^2 * f(0) + \left(1 - \frac{3}{2}\right)^2 * f(1) + \cdots = \frac{3}{4}$

Example 1

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- What is $E[X]$?
- $E[X] = 0 * f(0) + 1 * f(1) + 2 * f(2) + 3 * f(3) = \frac{3}{2}$
- What is $\mathrm{Var}[X]$?
- $\mathrm{Var}[X] = \left(0 - \frac{3}{2}\right)^2 * f(0) + \left(1 - \frac{3}{2}\right)^2 * f(1) + \cdots = \frac{3}{4}$
- Is there any easy way to compute $E[X]$ and $\mathrm{Var}[X]$?

Some Useful formulas

Formula 1

Linearity: $E[\sum_i X_i] = \sum_i E[X_i]$

$\boxed{\text{No assumption on } X_i}$

An Example

Toss a coin:

- If Head, you earn 2 dollar
- If Tail, you lose 1 dollar

Suppose you toss twice, how much you will earn on average?

Sample Space	HH	TT	HT	TH
Earnings: $\Sigma_i X_i$	4	-2	1	1
Probability	0.25	0.25	0.25	0.25

Average earning (mean): 4*0.25 - 2*0.25 + 1* 0.25 + 1*0.25 = 1$

An Example

Toss a coin:

- If Head, you earn 2 dollar
- If Tail, you lose 1 dollar

Suppose you toss n times, how much you will earn on average?

Sample Space	HH…H	…	…	TT…T
Earnings: $\Sigma_i X_i$	2n	…	…	-n
Probability	$\left(\frac{1}{2}\right)^n$	…	…	$\left(\frac{1}{2}\right)^n$

Difficult to calculate directly!

An Example

Toss a coin:

- If Head, you earn 2 dollar
- If Tail, you lose 1 dollar

Suppose you toss n times, how much you will earn on average?

$$\text{Linearity: } E[\Sigma_i X_i] = \Sigma_i E[X_i]$$

- For each toss, you win: 2*0.5-1*0.5=0.5
- In total, you win 0.5*n.

Much easier!

Hat Check

n people go to a party and leave their hat with a hat-check person. At the end of the party, she returns hats randomly since she doesn't care about her job. Let X be the number of people who get their original hat back. What is $E[X]$?

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Brute Force: $\Omega_X = \{0,1,2,\ldots,n-2,n\}$.

Sample space

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Brute Force: $\Omega_X = \{0,1,2,\ldots,n-2,n\}$.

$$p_X(n) = \frac{1}{n!}$$

$$p_X(0) = ???$$

Too hard → Use linearity!

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Quick question: does it matter where you are in line?

The key observation here is that the hats are returned **randomly** by the hat-check person. In other words, the hat-check person **randomly** selects a hat for each person, without any regard for who originally owned the hat. Since there are n hats in total, the hat-check person randomly selects one of the n hats to return to person $1/n$.

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If first in line, $P(\text{get hat back}) = \frac{1}{n}$, because there are n in total.

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Quick question: does it matter where you are in line?

If first in line, $P(\text{get hat back}) = \frac{1}{n}$, because there are n in total.
If last in line, $P(\text{get hat back}) = \frac{1}{n}$, because there is 1 left, and the chance it is yours is $\frac{1}{n}$.

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For $i = 1, \dots, n$, let $X_i = \begin{cases} 1, & \text{if } i^{th} \text{ person got hat back} \\ 0, & \text{otherwise} \end{cases}$. Then $X = \sum_{i=1}^{n} X_i$.

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$E[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) =$

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$$E[X] = E\left[\sum_{i=1}^{n} X_i\right] \overset{\text{Linearity}}{=} \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} \frac{1}{n} = n \cdot \frac{1}{n} = 1$$

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NOT independent Random variables (why?)

We will use linearity of expectation.

$$E[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = P(X_i = 1) = P(i^{th} \text{ person got hat back}) = \frac{1}{n}$$

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} \frac{1}{n} = n \cdot \frac{1}{n} = 1$$

Formula 2

$$\mathrm{E}[g(X)] = \Sigma_x \, g(x) P(X = x) = \Sigma_x \, g(x) f(x)$$

- Note: **$\mathrm{E}[g(X)] \neq g(\mathrm{E}[X])$**

- Toss a coin: head as 1, tail as -1.
- Then $\mathrm{E}[X^2] = 1^2 \times \frac{1}{2} + (-1)^2 \times \frac{1}{2} = 1$ $\boxed{g(x) = x^2}$
- But $(\mathrm{E}[X])^2 = 0$

Variance

$$\underbrace{E[(X - E[X])^2]}$$

$$X^2 - 2XE[X] + (E[X])^2$$

$$\Downarrow$$

$$E[X^2] \underbrace{- 2E[X]E[X] + (E[X])^2}_{-\,(E[X])^2}$$

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$$E[X^2] \underbrace{- 2E[X]E[X] + (E[X])^2}_{-\,(E[X])^2}$$

More useful

$$Var(X) = E[(X - E[X])^2] \boxed{= E[X^2] - (E[X])^2}$$

Variance (Example)

Let X be the outcome of a fair 6-sided die roll. What is $Var(X)$?

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$$Var(X) = E[X^2] - E[X]^2 = \frac{91}{6} - (3.5)^2 = \frac{35}{12}$$